



Housing Recovery and Reconstruction Platform-Nepal (HRRP)

Orientation Program for *Partner organizations*

on

Confined Masonry Construction

Organized by:

National Reconstruction Authority(NRA)

MOUD/District Project Implementation Unit (DLPIU)

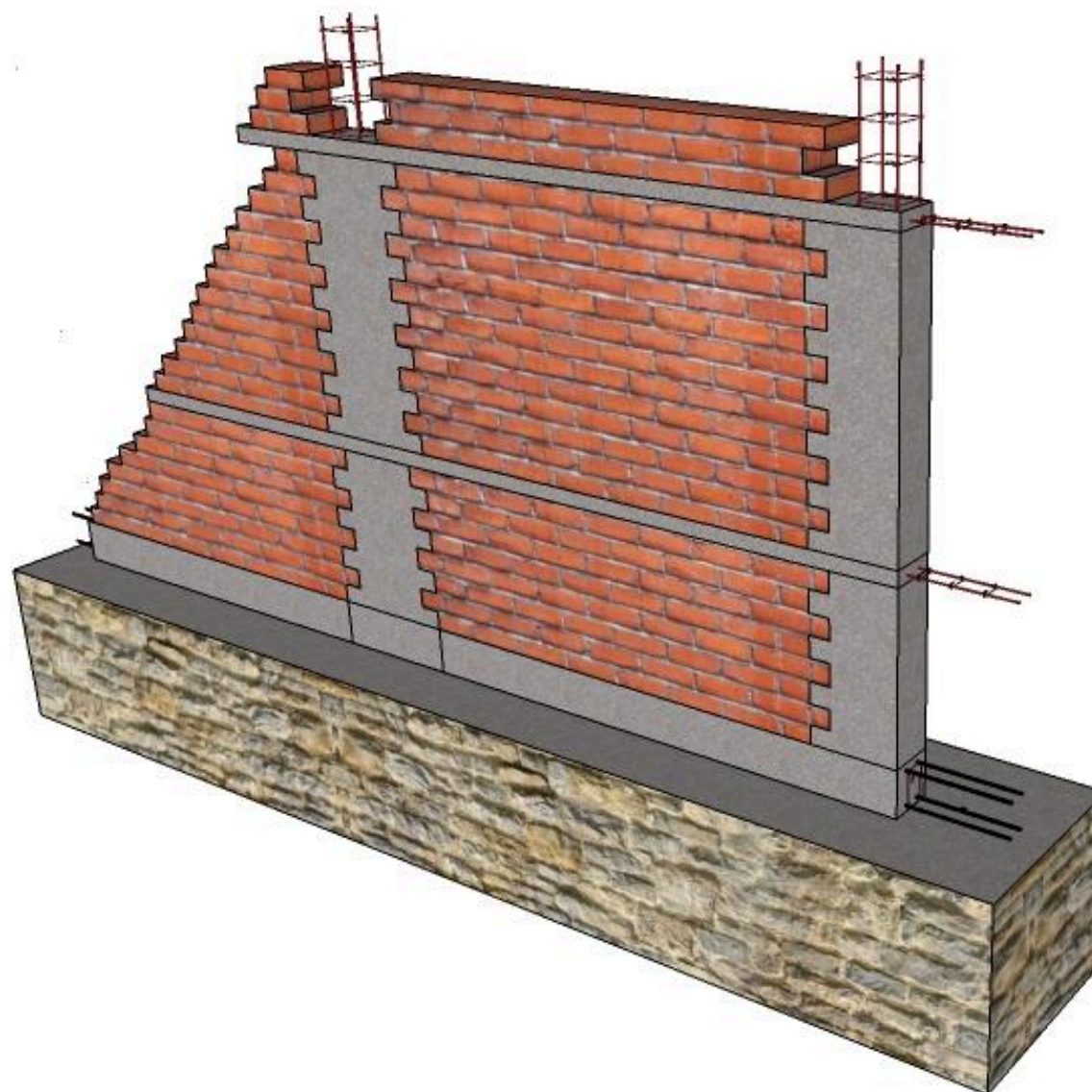
Housing Recovery and Reconstruction Platform-Nepal (HRRP)

Confined Masonry (CM) Construction



It consists of masonry walls and horizontal and vertical RC confining members built on all four sides of a masonry wall panel.





Why Confined Masonry ?



- Poor performance of unreinforced masonry and non ductile reinforced concrete (RC) frame construction caused unacceptably high human and economic losses in past earthquakes
- This prompted a need for developing and/or promoting alternative building technologies

The goal is to achieve enhanced seismic performance using technologies which require similar (preferably lower) level of construction skills and are economically viable

Why Confined Masonry in Nepal?



An opportunity for improved seismic performance both for unreinforced masonry and reinforced concrete frame construction in low- and medium-rise buildings

An Alternative to RC Frame Construction



An Alternative to Unreinforced Masonry Construction



Confined Masonry and RC Frame Construction: Performance in Recent Earthquakes



January 2010, Haiti

M 7.0

300,000 deaths



February 2010, Chile

M 8.8 521 deaths

**(10 due to confined masonry
construction)**

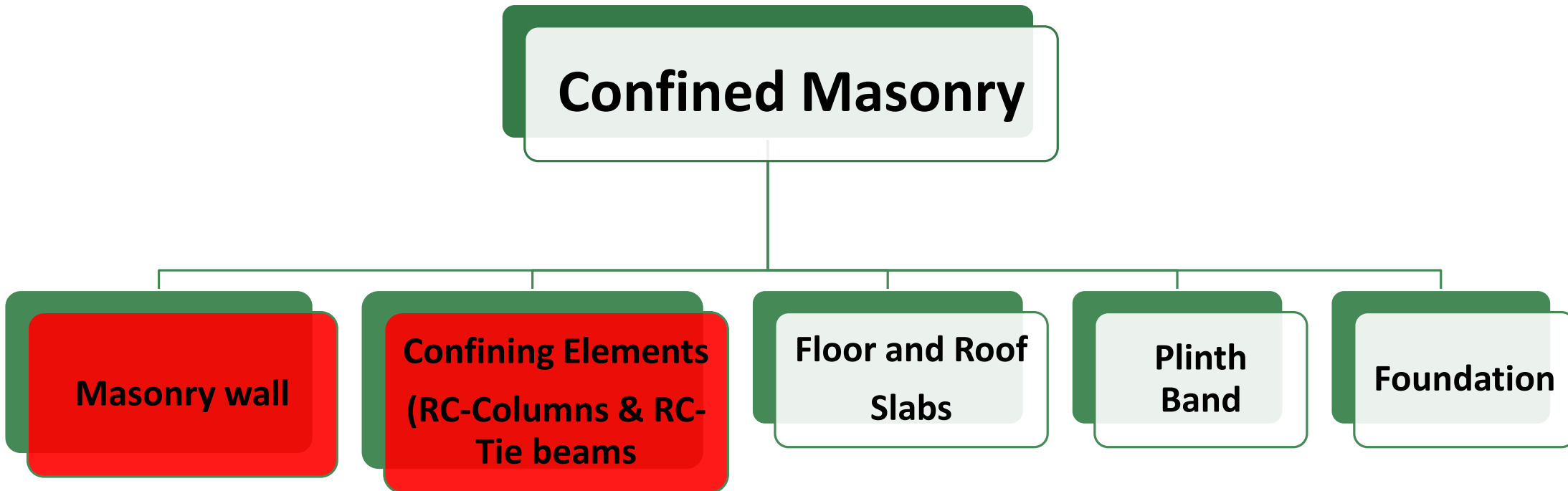
Beginnings



- Evolved through an informal process based on its satisfactory performance in past earthquakes
- The first reported use in the reconstruction after the 1908 Messina, Italy earthquake (M 7.2) - death toll 70,000
- Practiced in Chile and Columbia since 1930's and in Mexico since 1940's

Currently practiced in several countries/regions with high seismic risk, including Latin America, Mediterranean Europe, Middle East (Iran), South Asia (Indonesia), and the Far East (China).

Components of CM

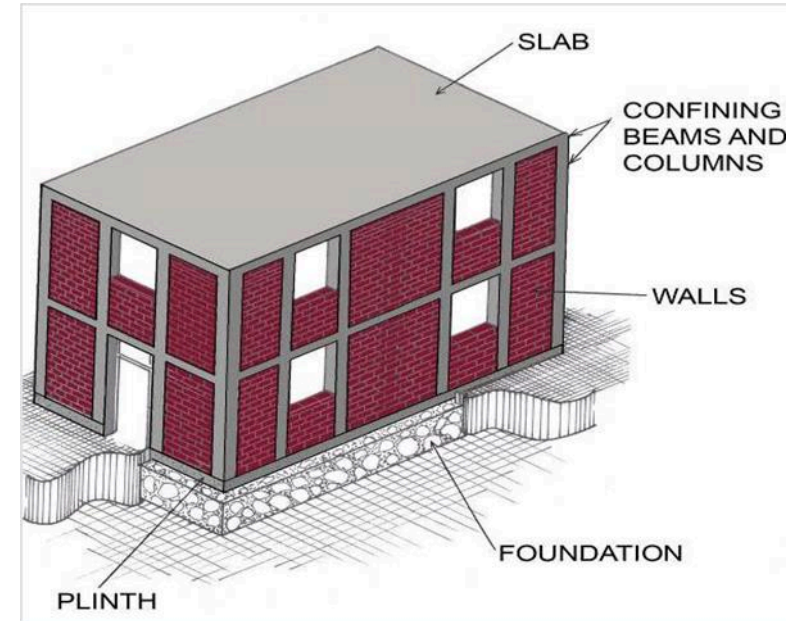


Key features of Structural Component

Confining elements (RC tie-columns and RC tie-beams)



- Horizontal(tie-beams) & Vertical(tie-columns) RC confining members built in all four side of wall
- Provide restraint to masonry walls and protect them from complete disintegration even in major earthquakes
- Enhance the stability and integrity of masonry walls for in-plane & out- of- plane earthquake loads.



Confining elements (RC tie-columns and RC tie-beams)



- Enhance the strength(resistance) of masonry wall under lateral earthquake loads
- Reduce the brittleness of masonry walls under earthquake hence improving their earthquake performance



Masonry Wall



- Made of solid clay bricks, hollow clay tiles, or concrete blocks
- Confined by tie-columns & tie-beams
- Transmit the gravity load from the slab(s) above down to the foundation
- Act as bracing panels, which resist horizontal earthquake forces

Floor & roof Slabs



- Transmit both gravity & lateral loads to the wall
- In Earthquake slab behave like a horizontal beam & Called diaphragm

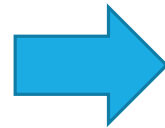
Plinth band

- Transmit the load from wall to the foundation

Foundation

- Transmit the load from the structure to the ground

Confined Masonry versus Reinforced Masonry



Vertical and horizontal reinforcing bars are provided to enhance the strength and ductility of masonry walls

Reinforcement is concentrated in vertical and horizontal RC confining elements whereas the masonry walls are usually free of reinforcement

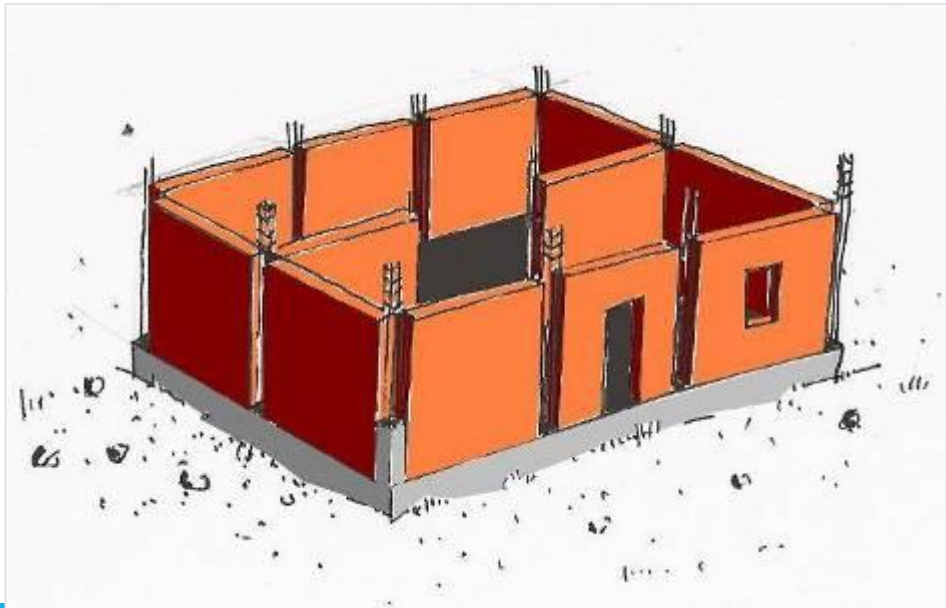
Confined Masonry versus Infilled RC frames:



- construction sequence
- integrity between masonry and frame

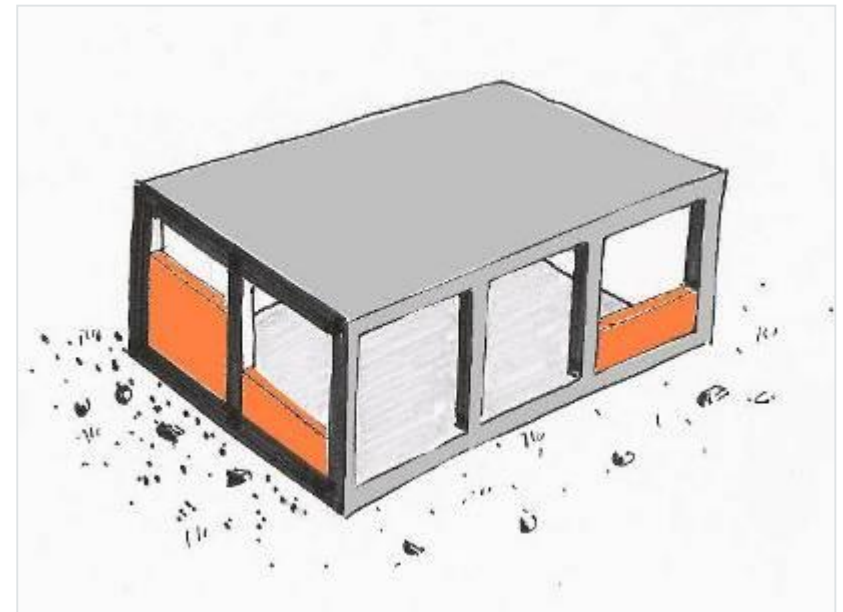
Confined Masonry

- Walls first
- Concrete later

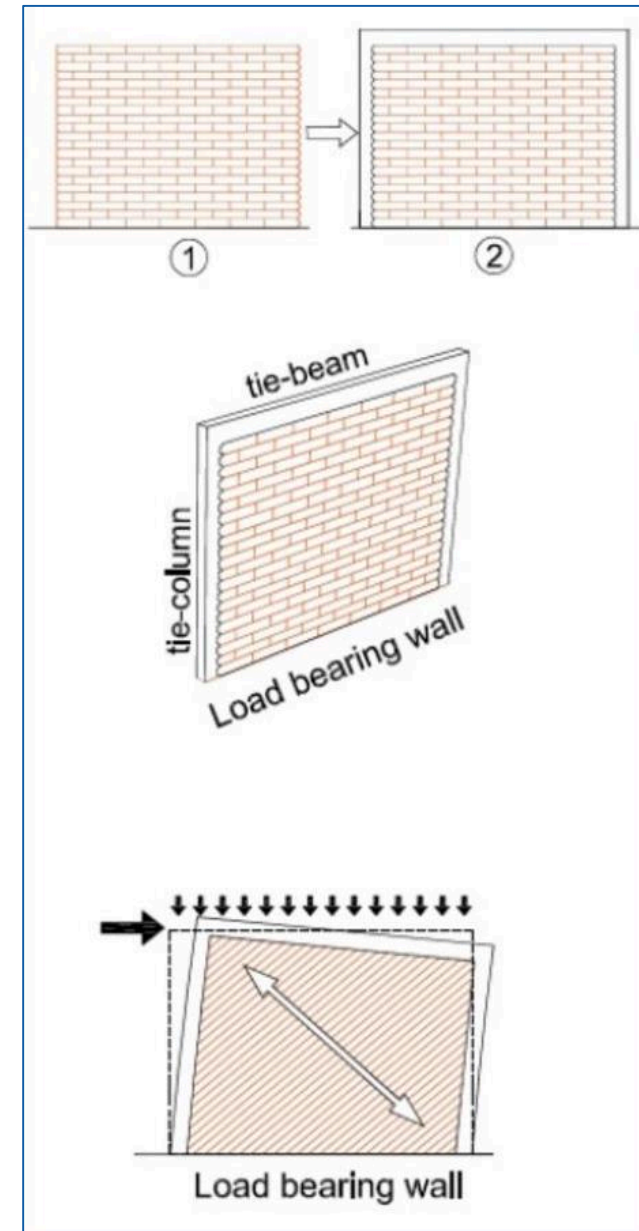
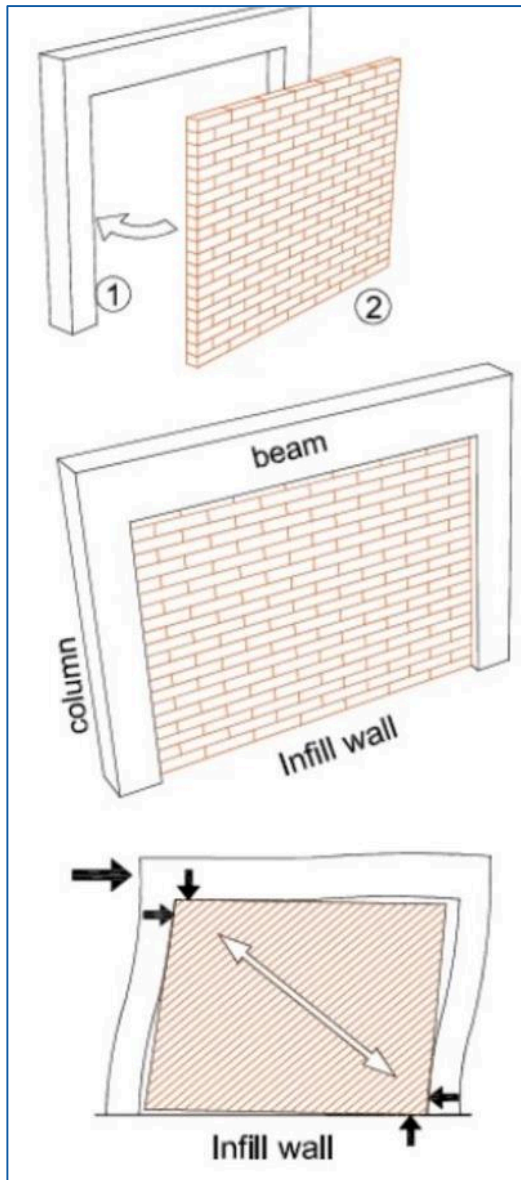


Reinforced Concrete Infilled Frame

- Concrete first
- Walls later



Confined Masonry vs RC Frames with Infills – Key Differences



Failure modes of Confined Masonry buildings (not well designed or constructed)



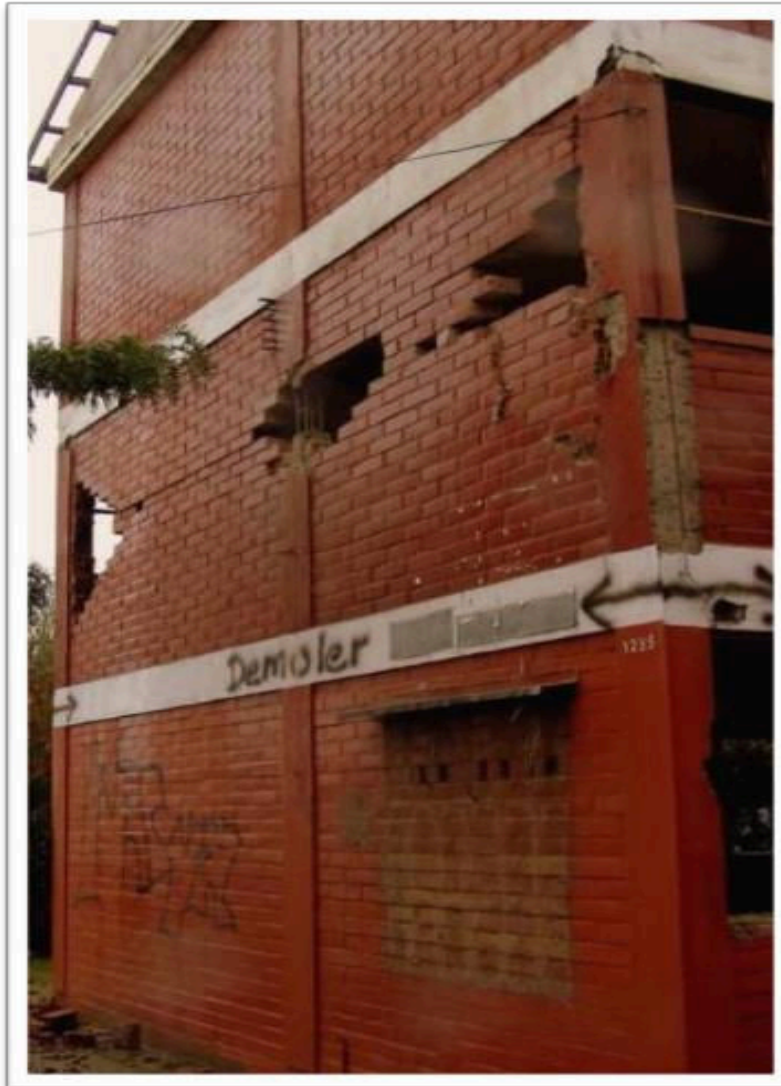
- Masonry damage (in- and out-of-plane)
- RC tie-columns
- Tie-beam-to-tie-column joints
- Confining elements around openings

In-plane shear failure of masonry wall



Earthquake ground shaking in the direction parallel with the longitudinal wall axis

Out-of-Plane Wall Damage



- An example of out-of-plane damage observed in a three-storey building
- The damage concentrated at the upper floor levels
- The building had concrete floors and timber truss roof
- The same building suffered severe in-plane damage

Earthquake ground shaking perpendicular to the longitudinal wall axis

Out-of-Plane Wall Failure

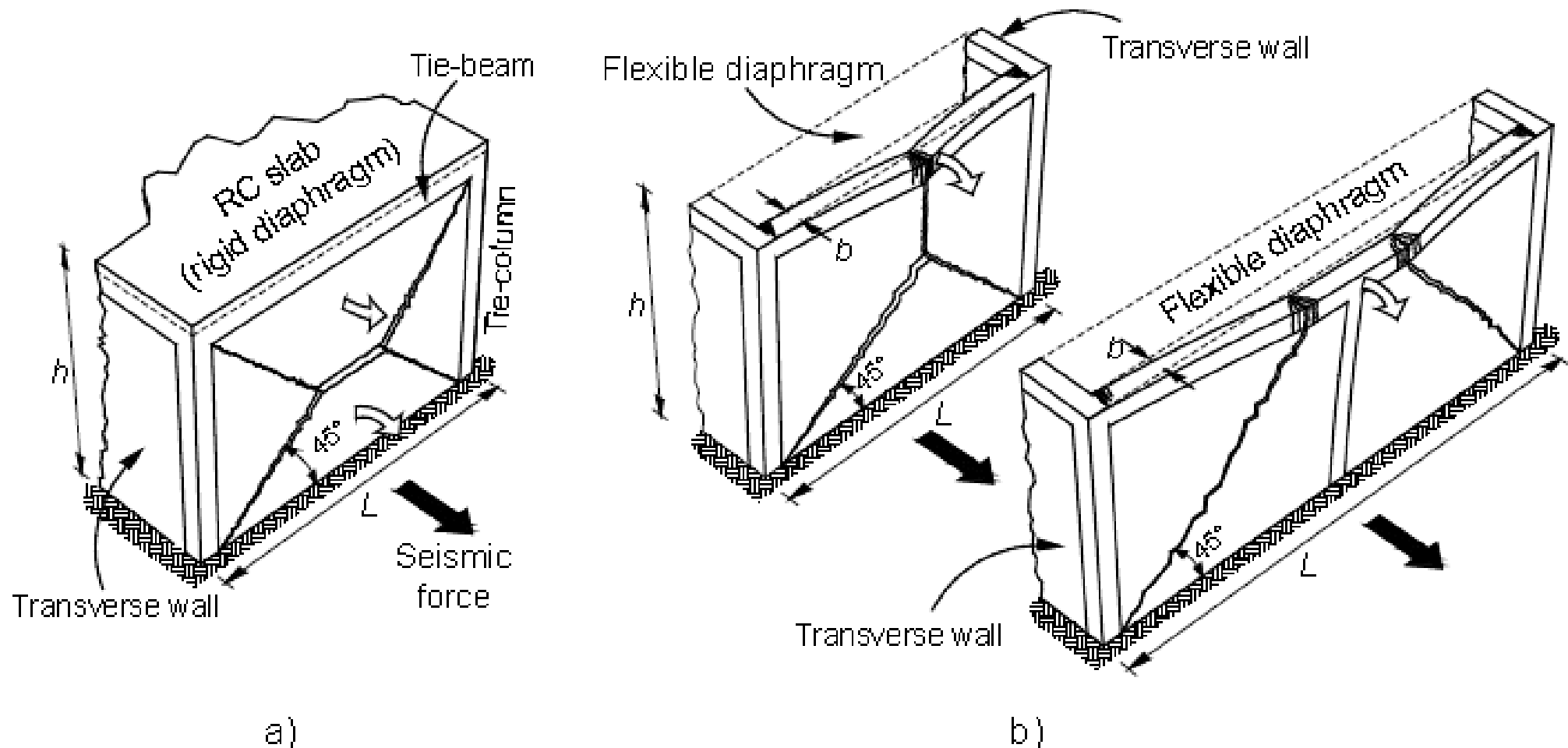
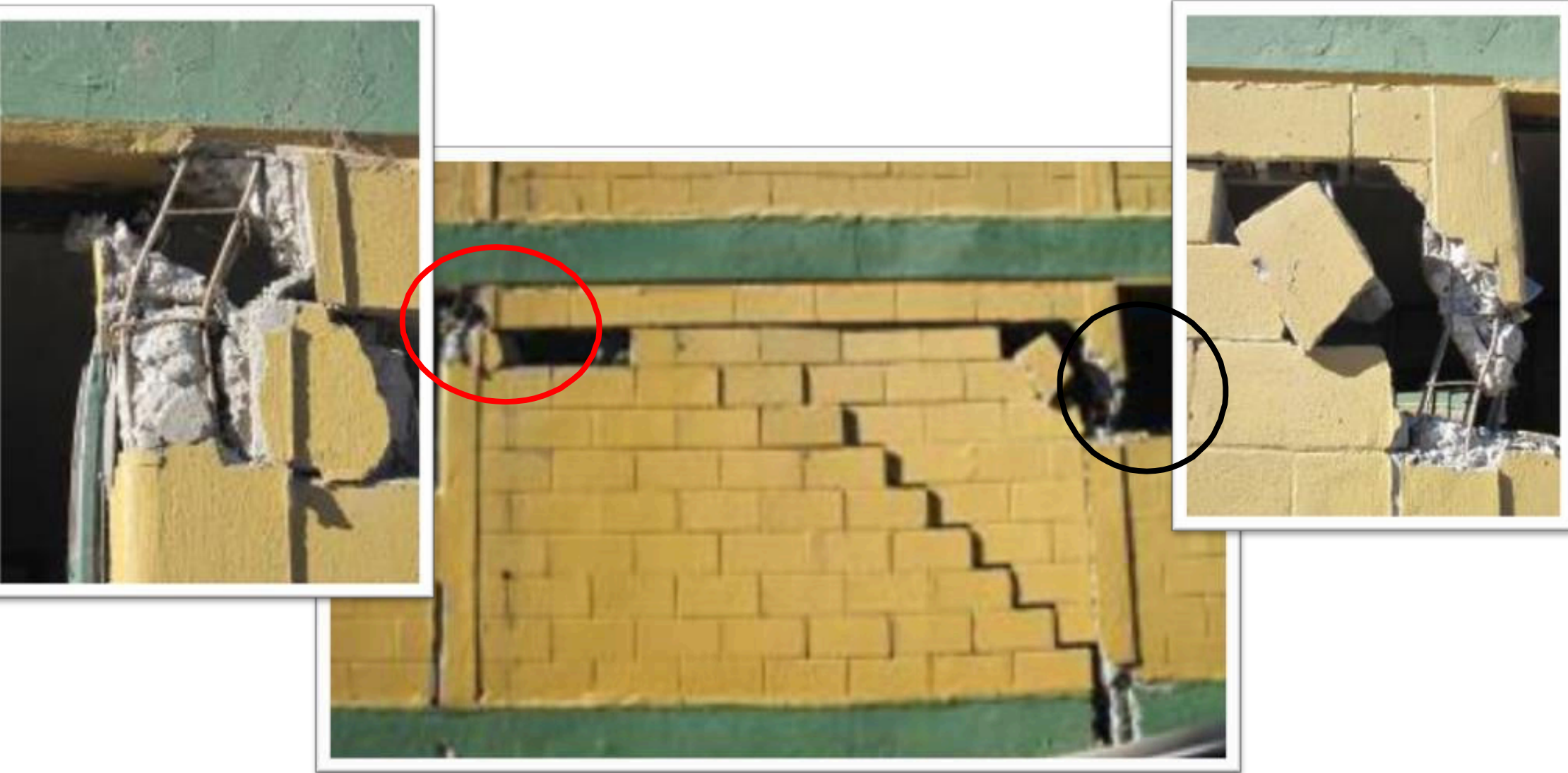
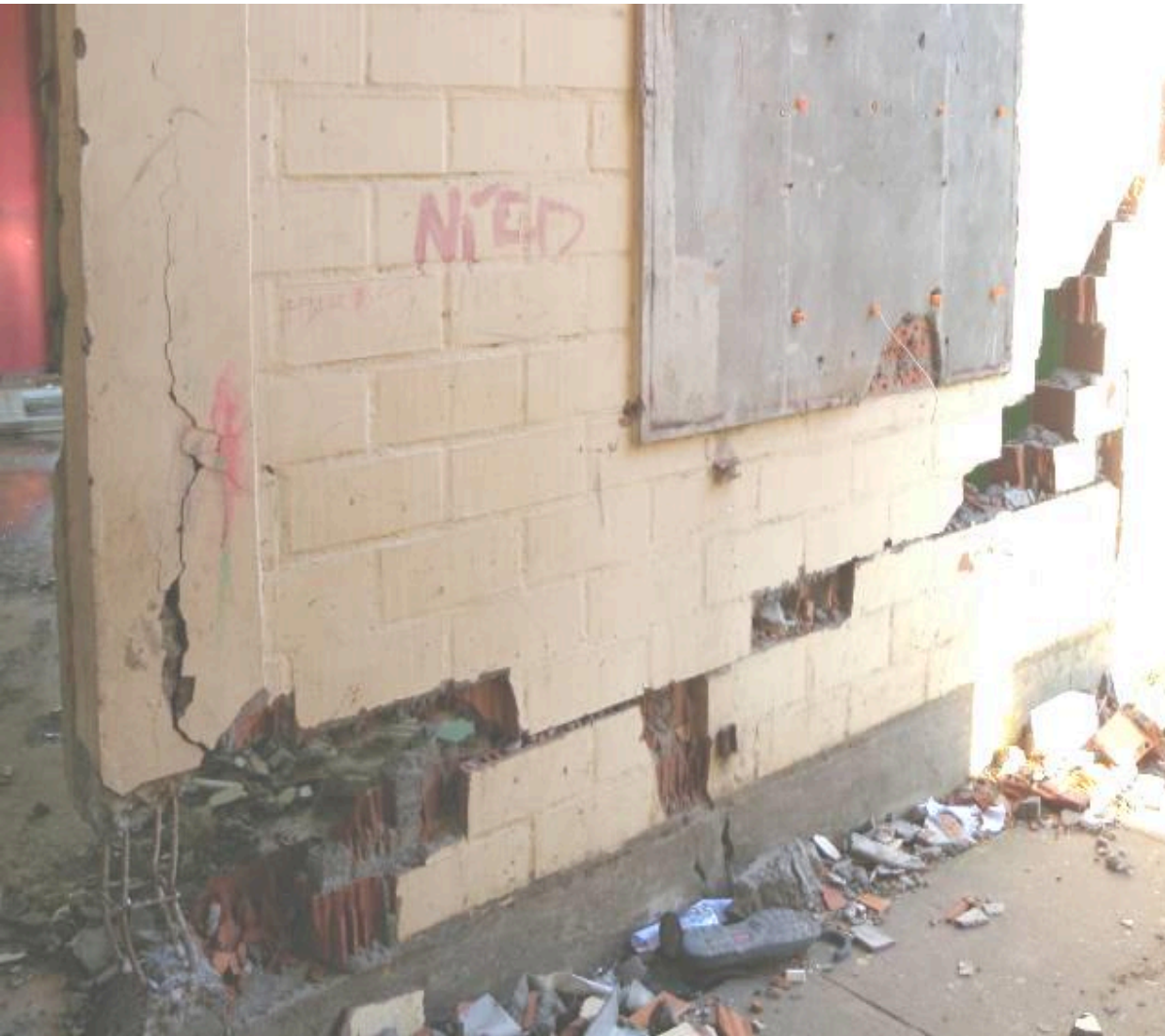


Figure 47. Mechanisms of failure for confined masonry walls under the out-of-plane seismic loads: a) buildings with rigid diaphragms, and b) buildings with flexible diaphragms.

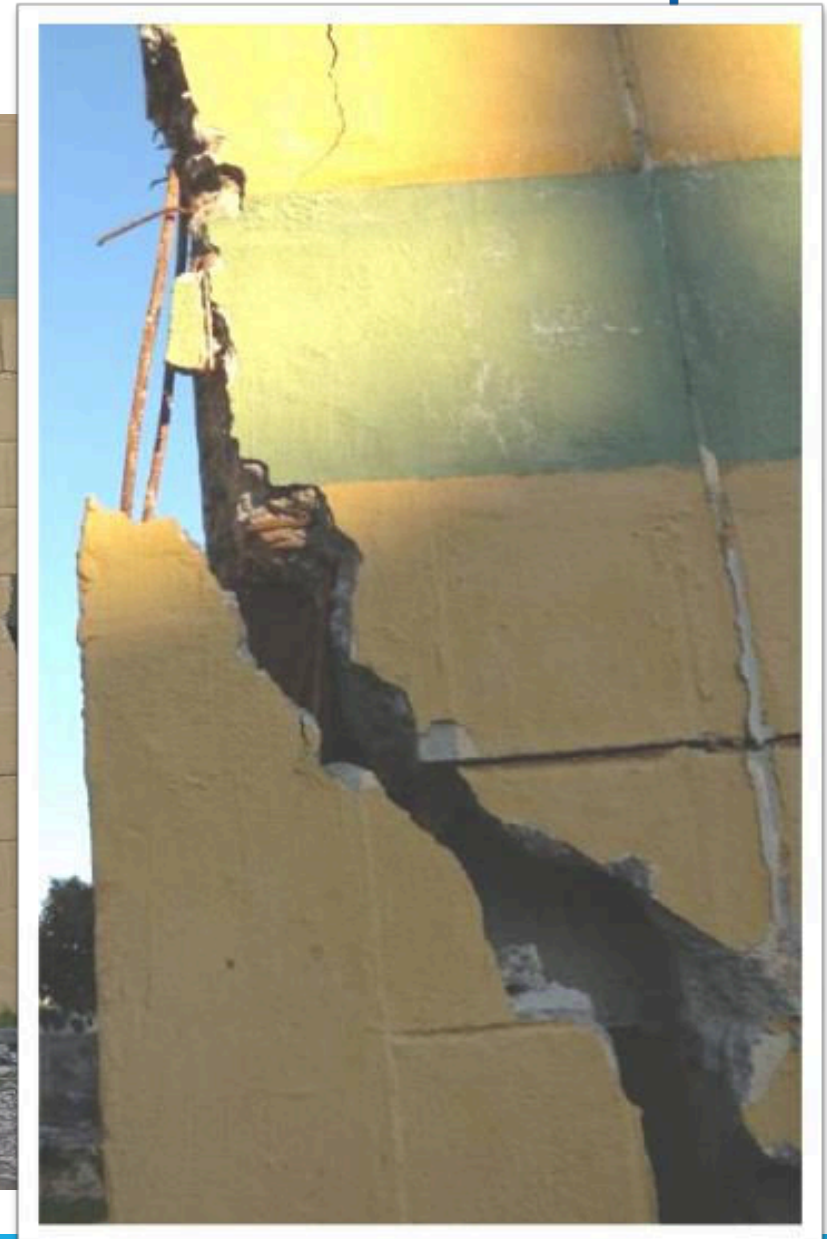
Tie-Column Failure



Buckling of a RC Tie-Column



Shear Failure of RC Tie-Columns

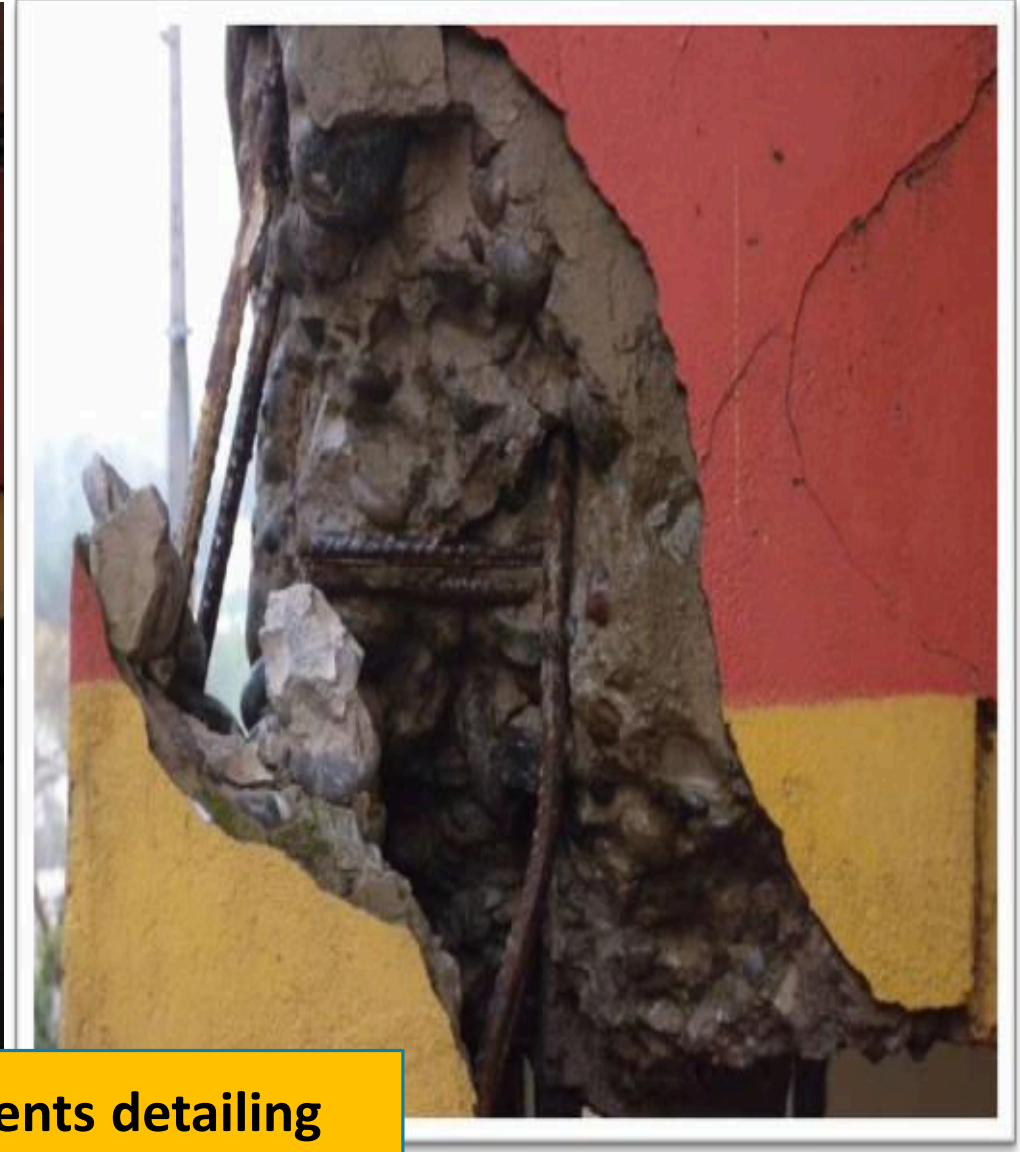


Tie-beam-to-tie-column joints



Inadequate anchorage of tie-beam reinforcement

Tie-beam-to-tie-column joints



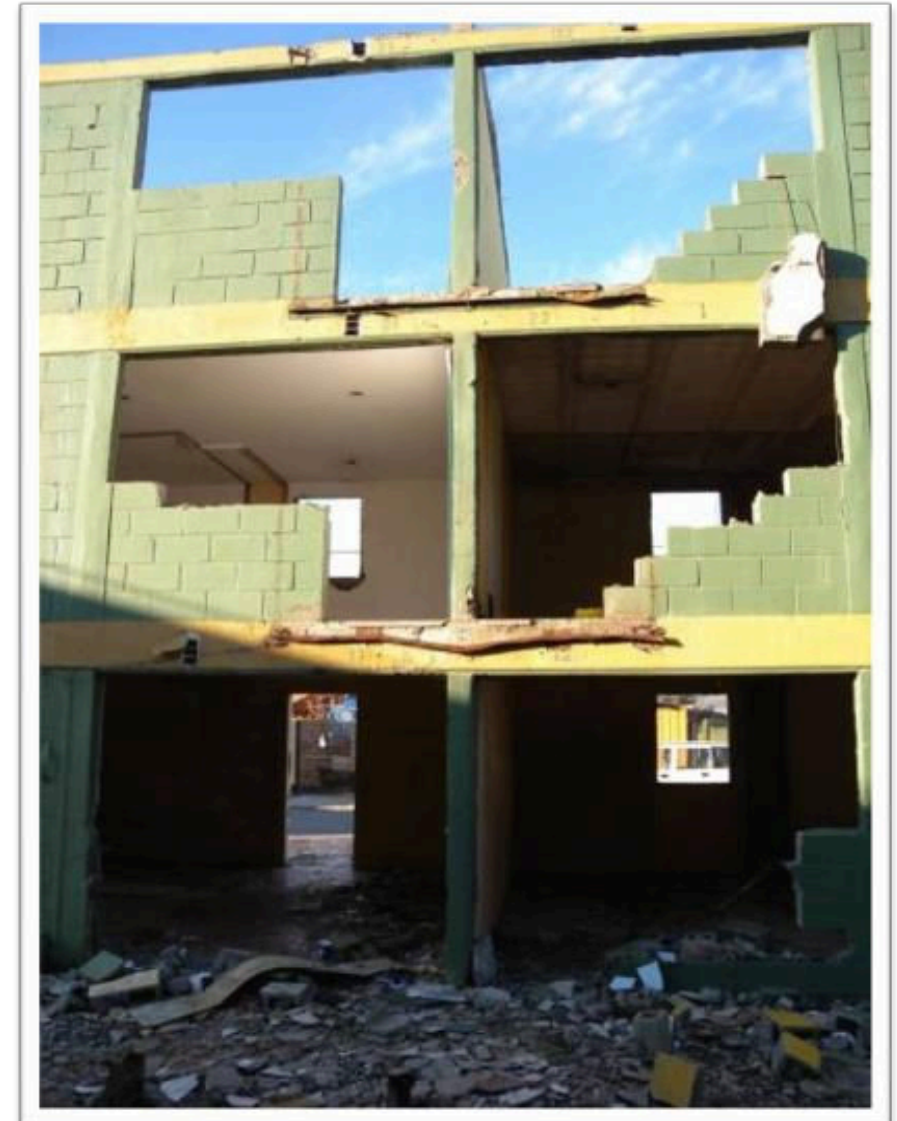
Due to Poor reinforcements detailing

Tie-beam-to-tie-column joints



Absence of of ties in joint Area

Absence of Confining elements around openings



In-Plane Shear Cracking – the Effect of Confinement



Unconfined openings



Confined openings

Key Causes of Damages



- Inadequate wall density
- Poor quality of masonry materials and construction
- Inadequate detailing of reinforcement in confining elements
- Absence of confining elements at openings

Construction Guideline for CM Construction



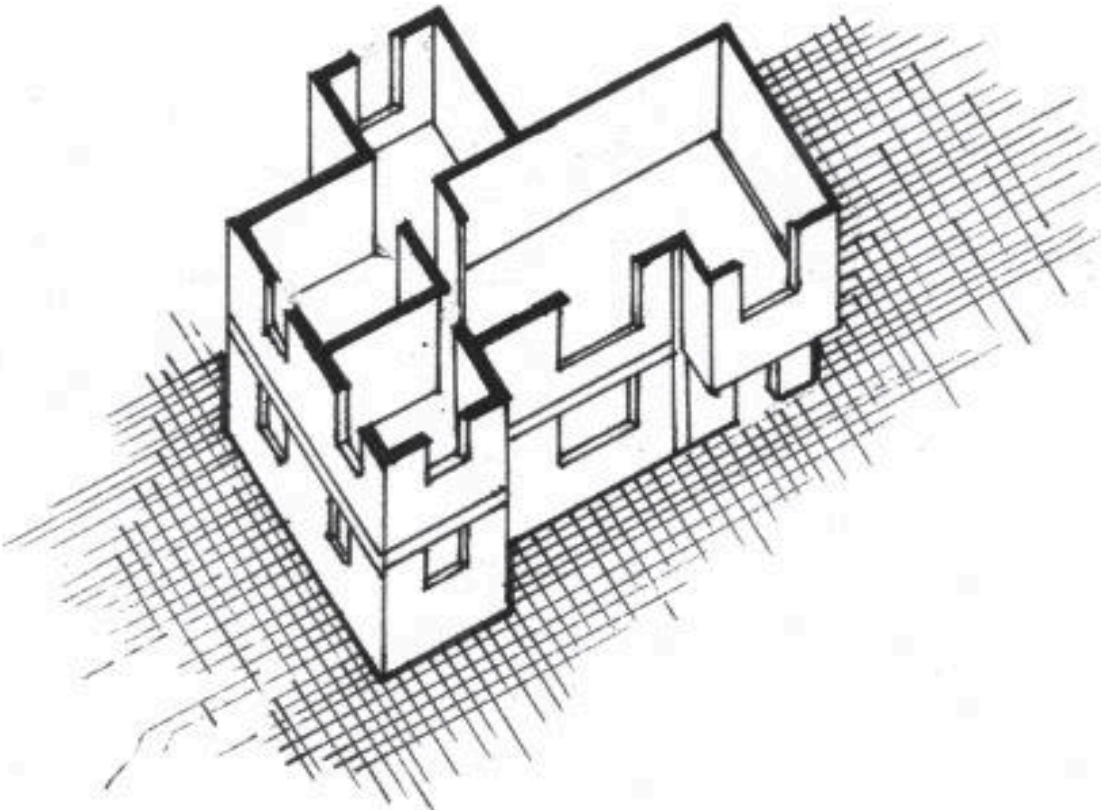
**Non-engineered Building
(Inspection form is for One storey only)**

- **Architectural Guideline**
- **Construction Guideline**

Architectural Guideline

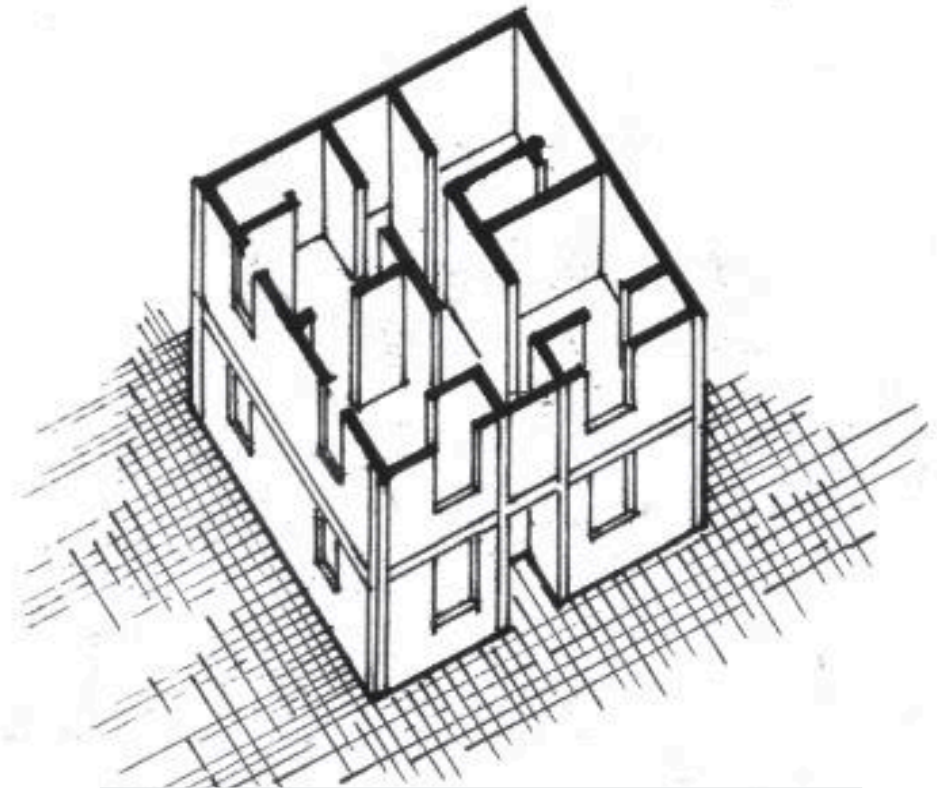
Symmetrical wall

X



Inadequate plan : layout

✓

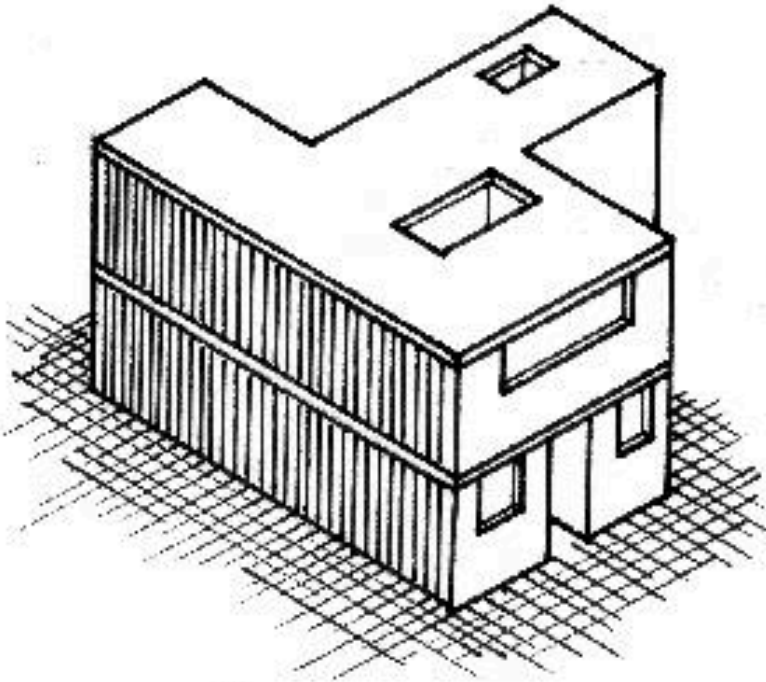


Adequate shape

Regular Building Plan

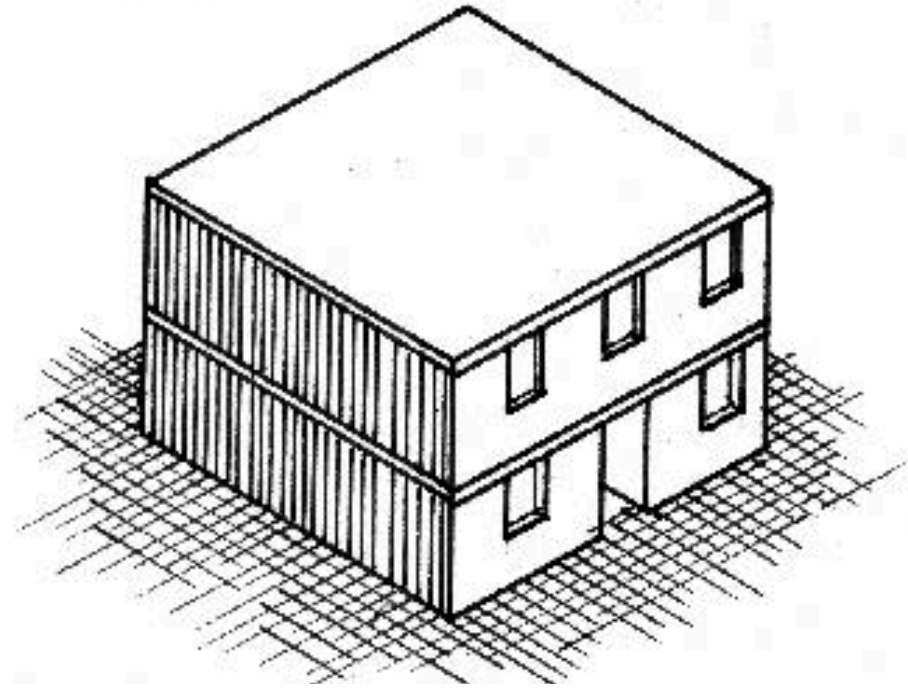


X



Irregular

✓



Symmetrical

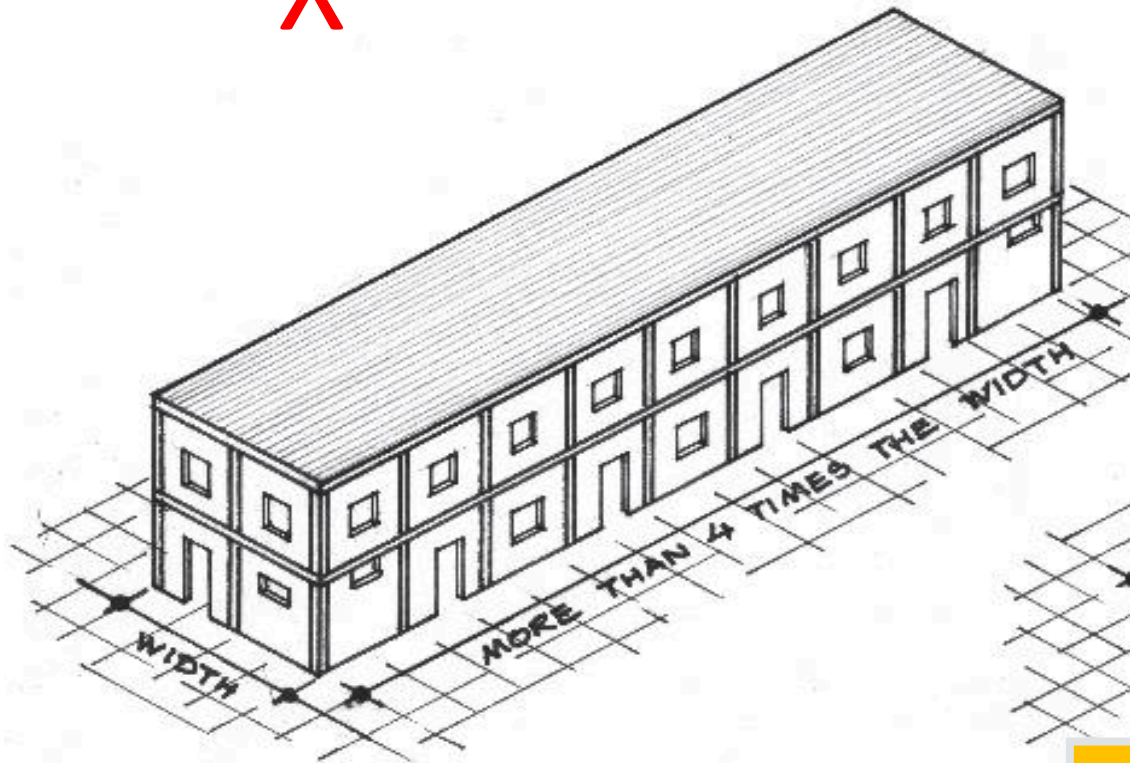
Plan Shape :

Length-to-width ratio less than 3 times

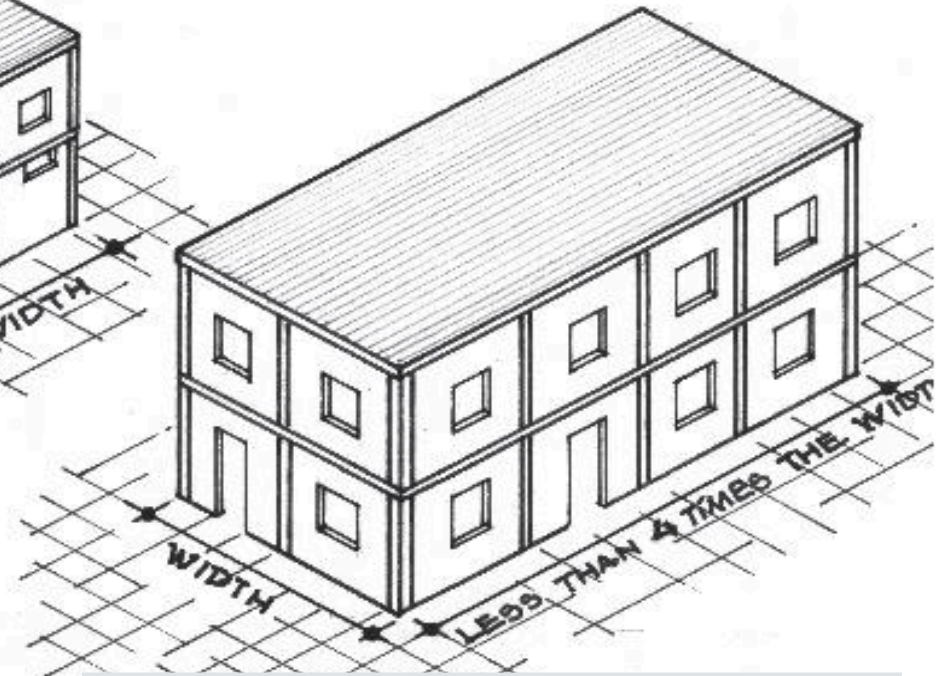


X

✓



Poorly proportioned plan

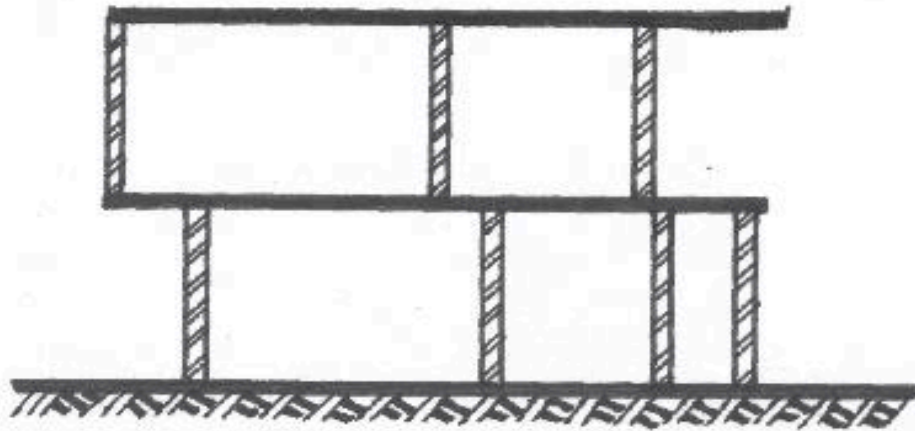


Well proportioned plan

Walls should be continuous up the building height

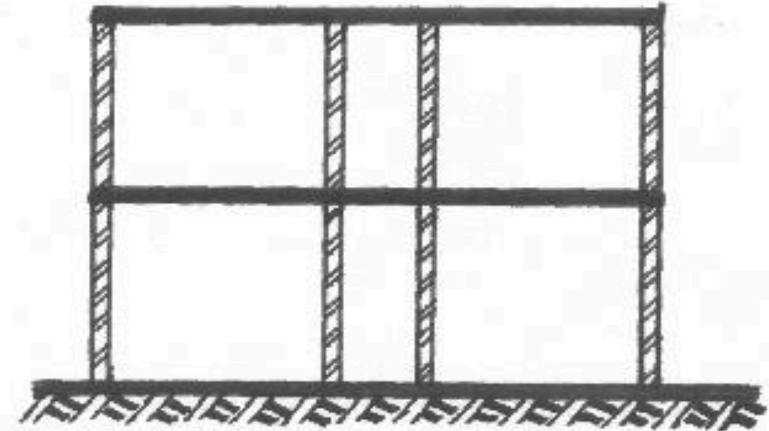


X



Load path not clear

✓

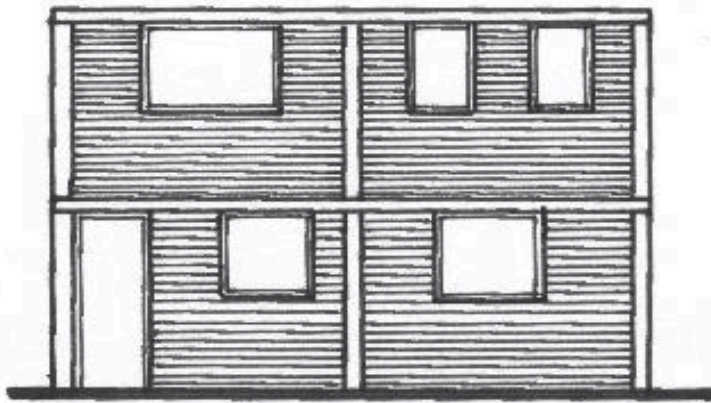


Load path clear

Wall height: 3m for Non-Designed Building.

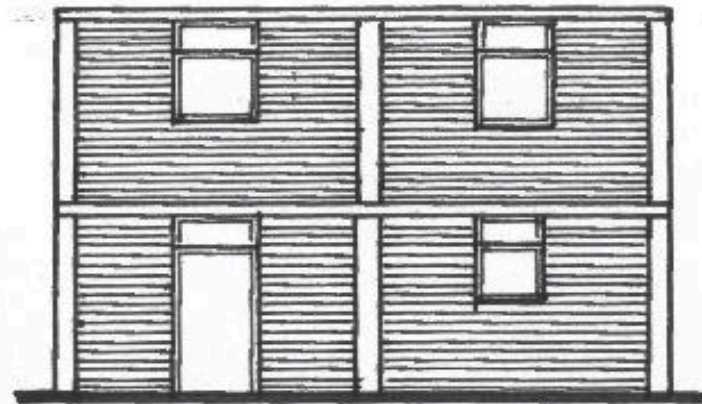
Opening

X



Poor Location of Window and door opening

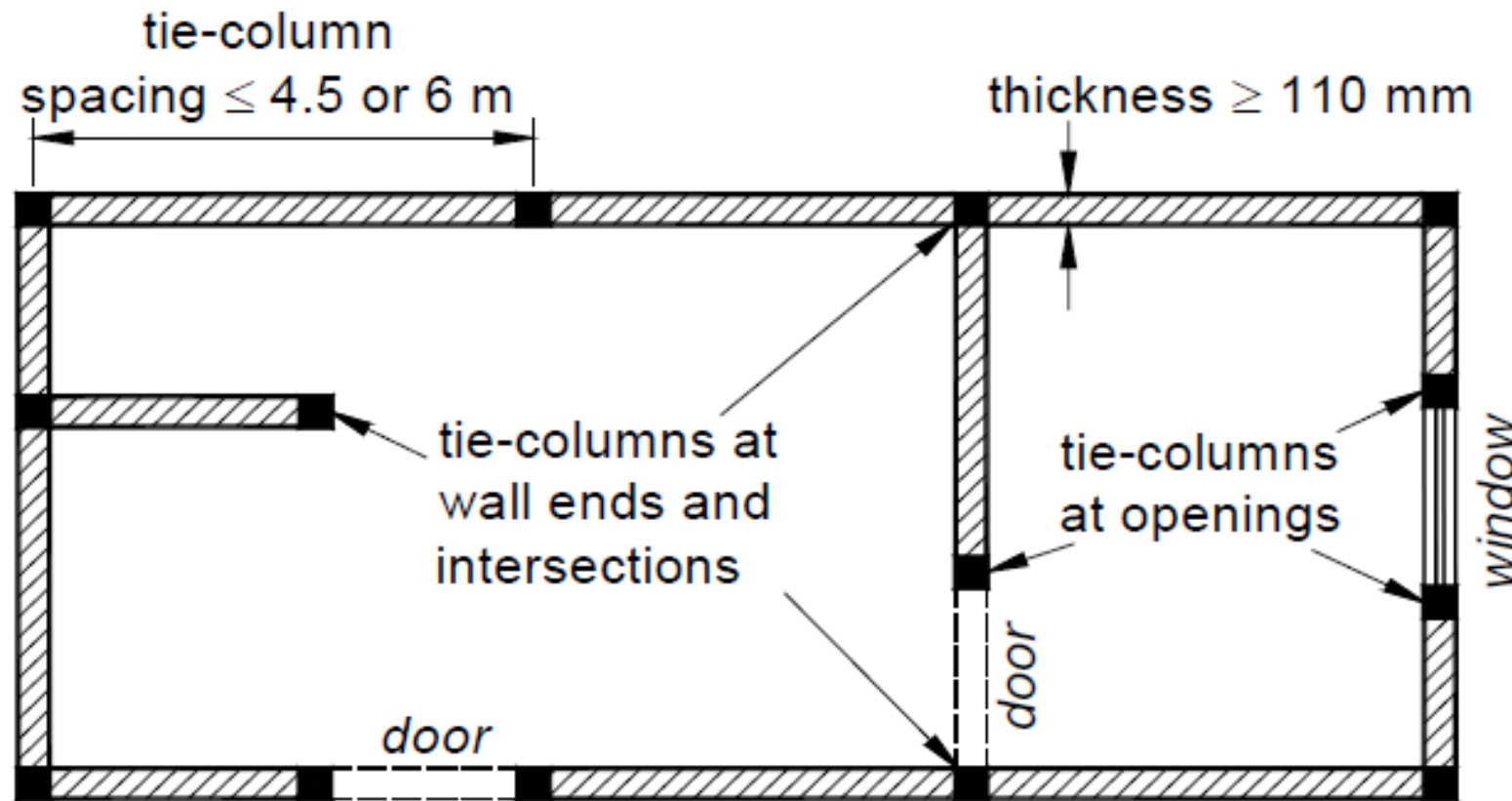
✓



Good location of window and door opening

Opening Should be placed in the same same position up the building height

Location of Confining Elements

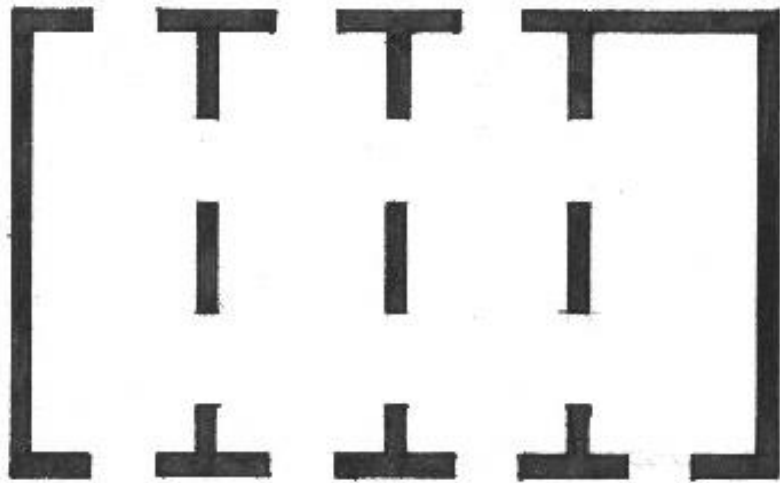


➤ floor every

- Tie-columns should be placed at a maximum spacing of less than 4.5 at wall to wall intersections , free end of wall & within wall if its lengths exceed than MRS

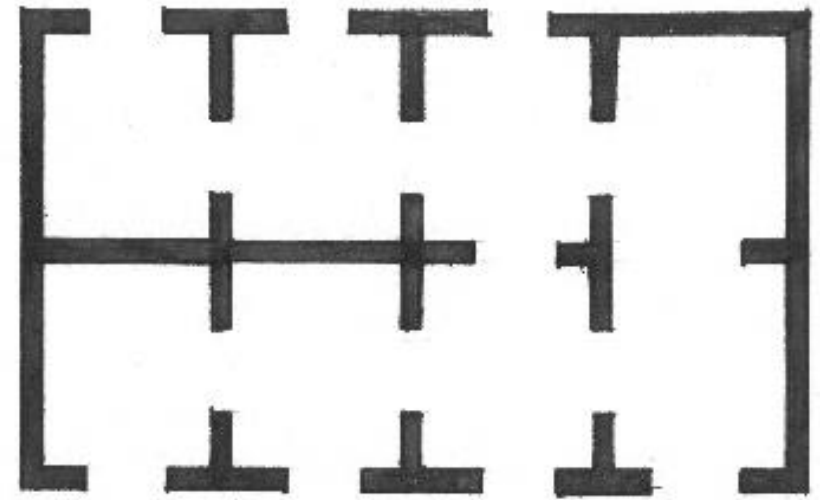
Wall Distribution

X



Inadequate Wall Distribution

✓



Adequate Wall Distribution

At least three fully confined walls should be provided in each direction

Walls Density



- At least 5 % in each of two orthogonal direction

$$\text{Wall Density} = \frac{\text{Total X – sectional Area of all walls in one direction}}{\text{Sum of Floor plan area for all floors in a building}}$$

Eurocode 8 (1996)

- a) At least 2% for a site with a design ground accln up to 0.2g (corresponding to seismic zone II of India)
- b) At least 4% for a site with a design ground accln up to 0.3g (corresponding to seismic zone III of India)
- c) At least 5% for a site with a design ground accln up to 0.4g (corresponding to seismic zone IV of India)

Construction Guideline

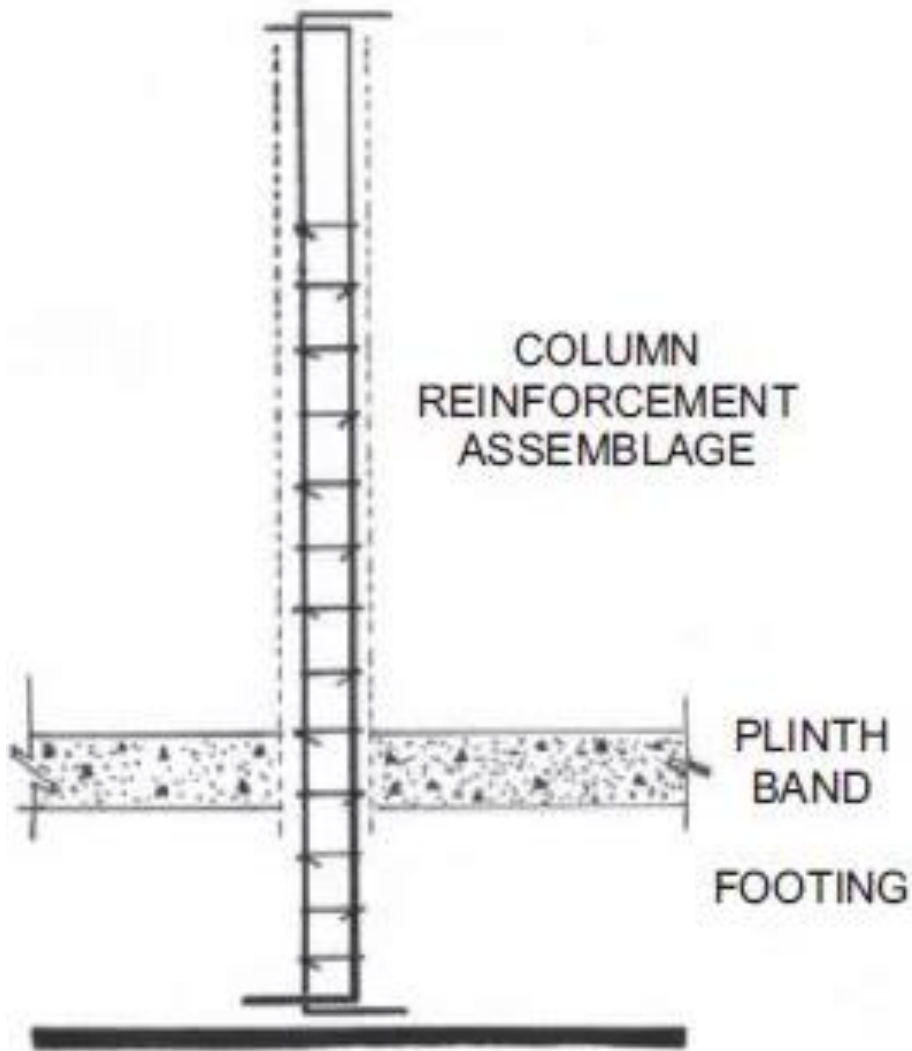
Wall (Constructing method)

Wall thickness: 230 mm



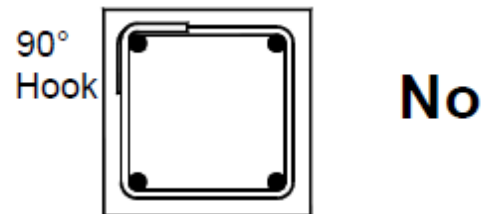
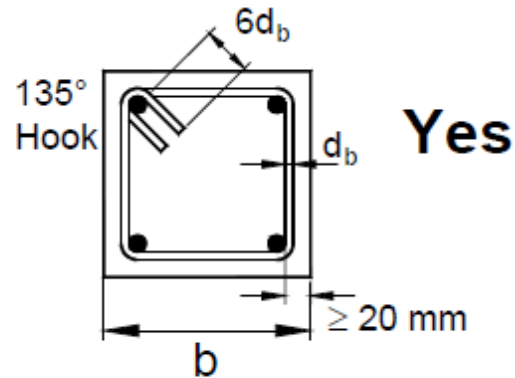
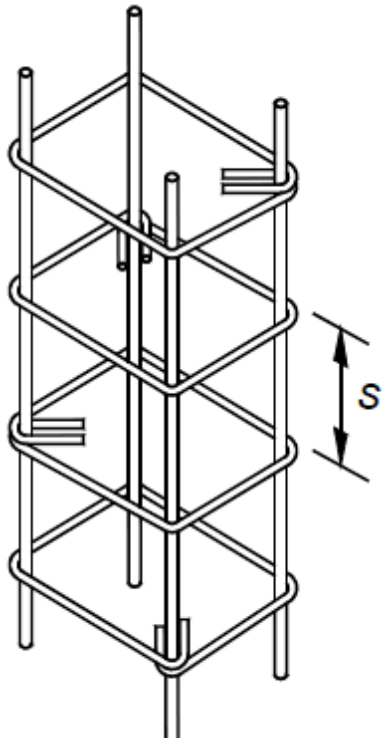
Toothing enhances interaction between masonry walls and RC confining elements

Tie -Column



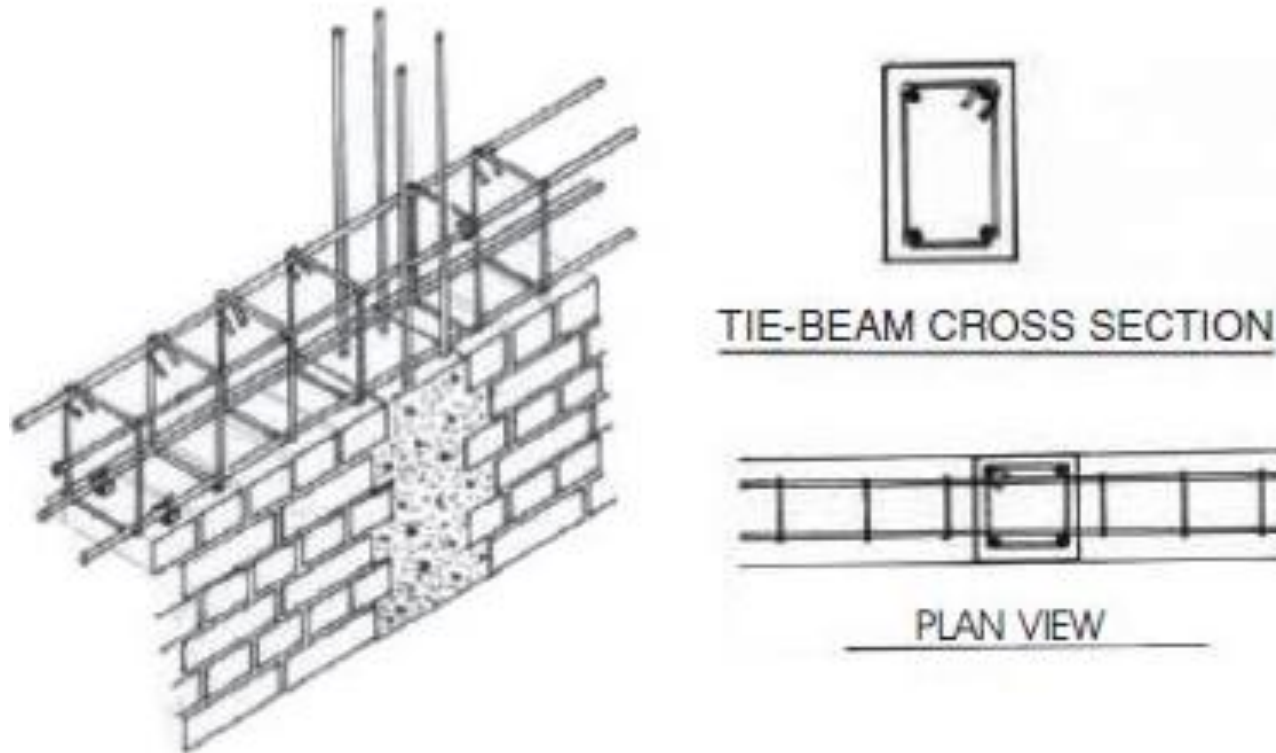
- Tie-column reinforcement set in the position prior to the Construction of Foundation and wall
- Should be Place at wall ends, intersection & Opening

Tie -Column (Reinforcements)



- **Tie Column Size :** (230 X 230)mm at Corner & wall Section
230 X 115) mm at Opening
- **Longitudinal reinforcements:** 4 ϕ 12mm bar at corner & wall section
2 ϕ 12mm bars at Opening
- **Stirrups:** 6 mm @ 150 c/c

Tie-beam

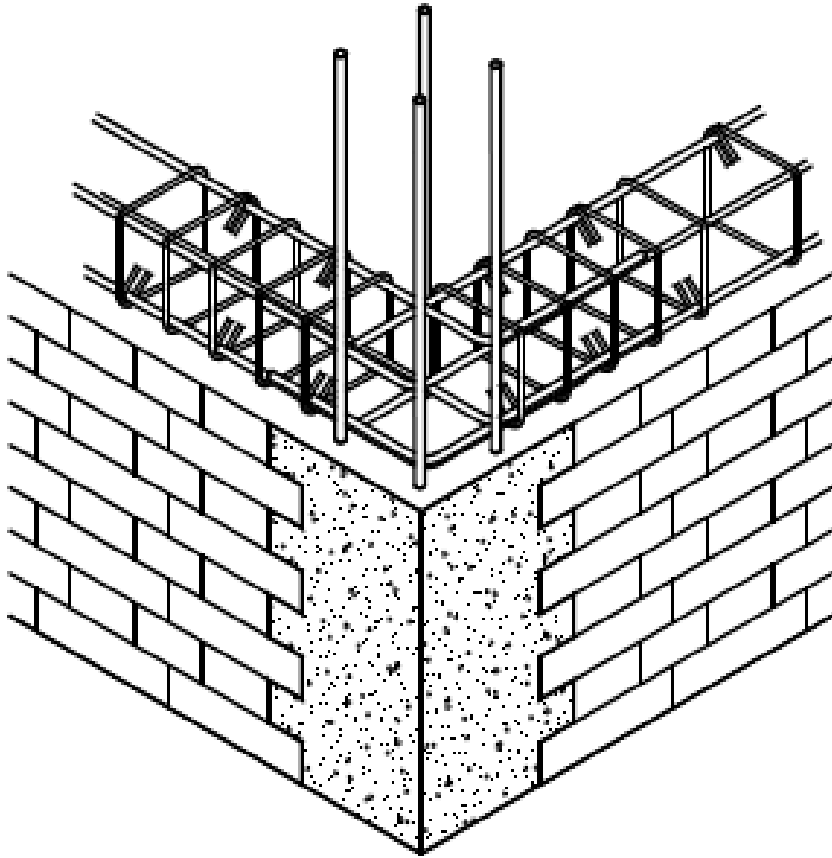


Tie-Beam are constructed a top the wall of each floor level

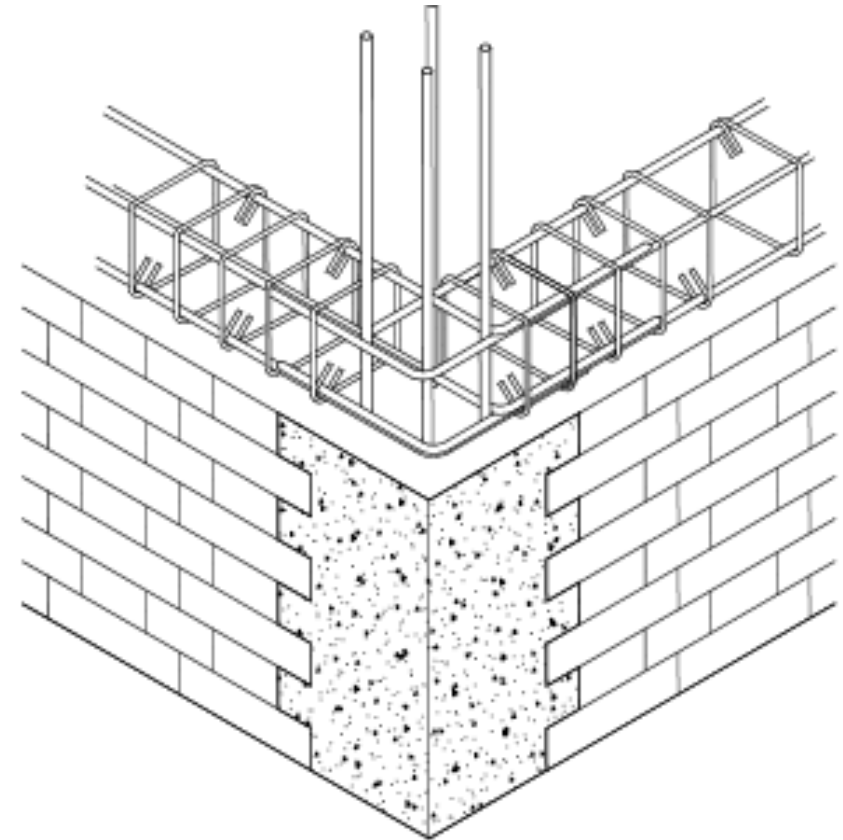
- **Horizontal ties Size :** (230 X 150)mm
- **Longitudinal reinforcements:** 4 ϕ 12mm bar at corner & wall section
- **Stirrups:** 6 mm @ 150 c/c

Joint Detailing

X



✓

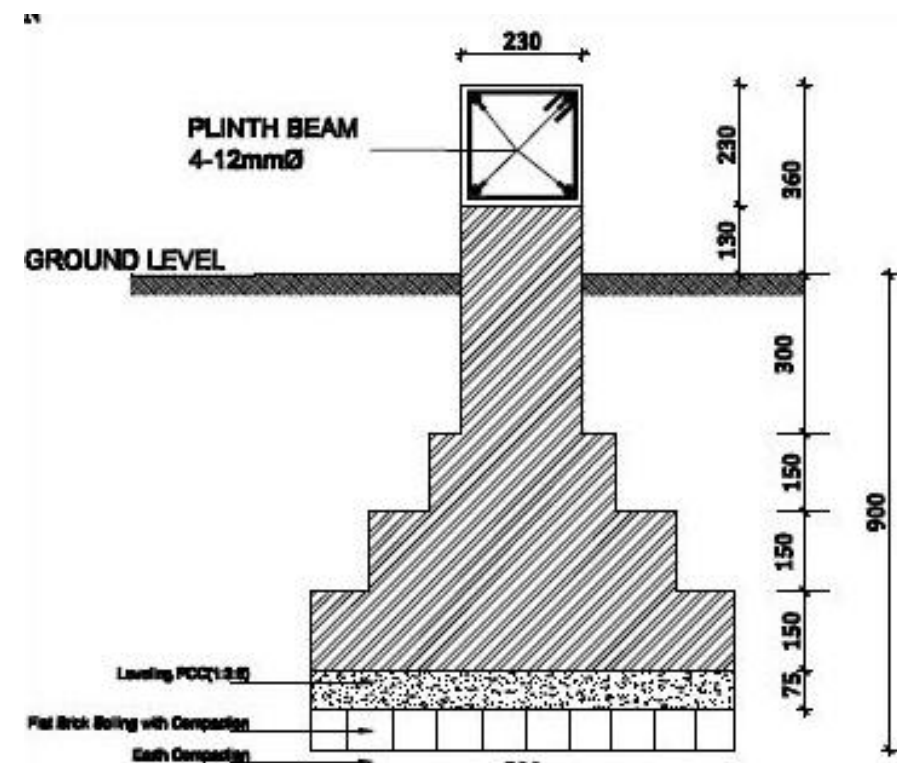
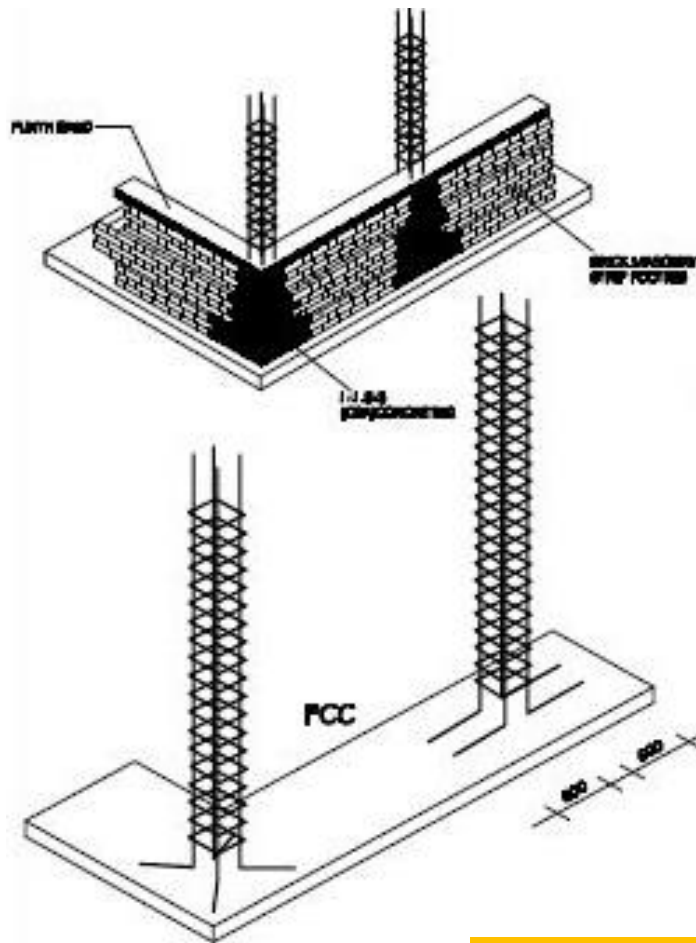


It is preferred to place beam reinforcement outside the column reinforcement cage

Foundation and Plinth Construction

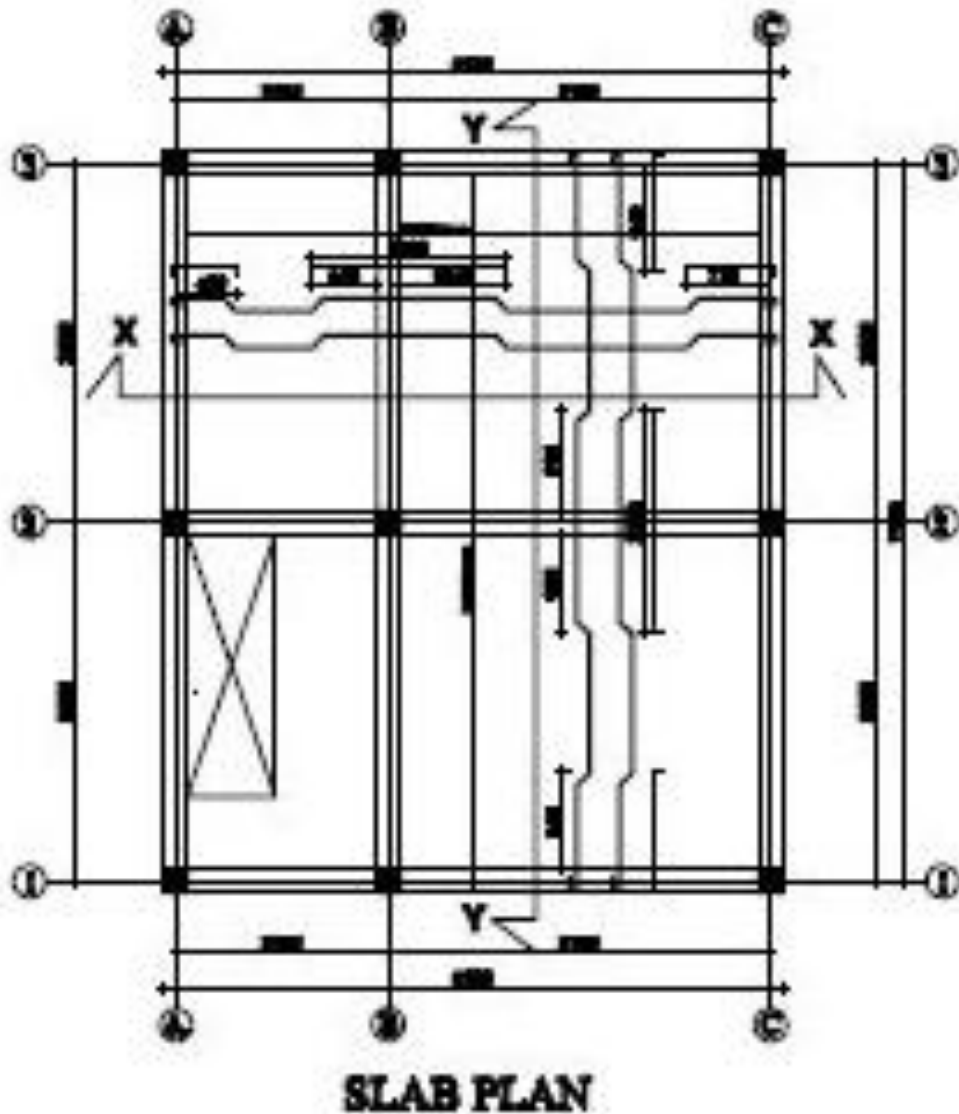


Similar as traditional masonry construction



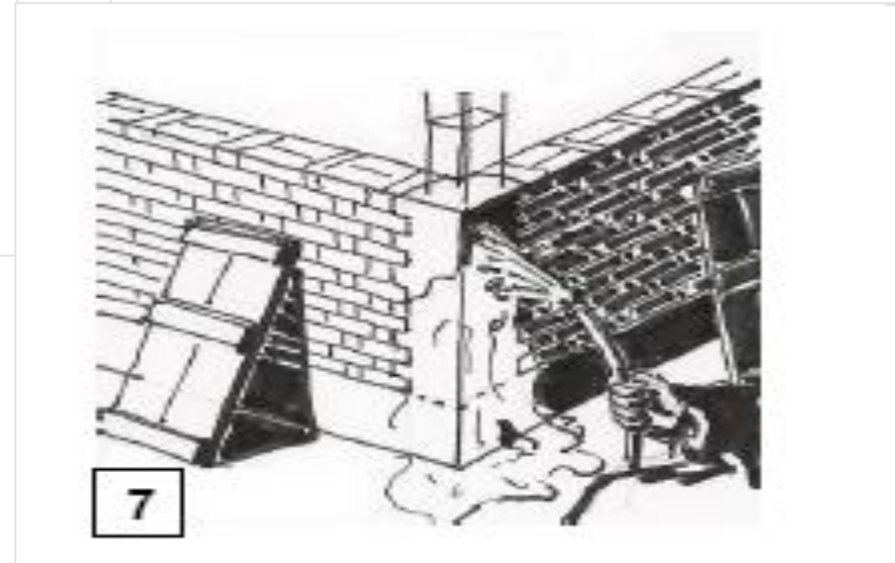
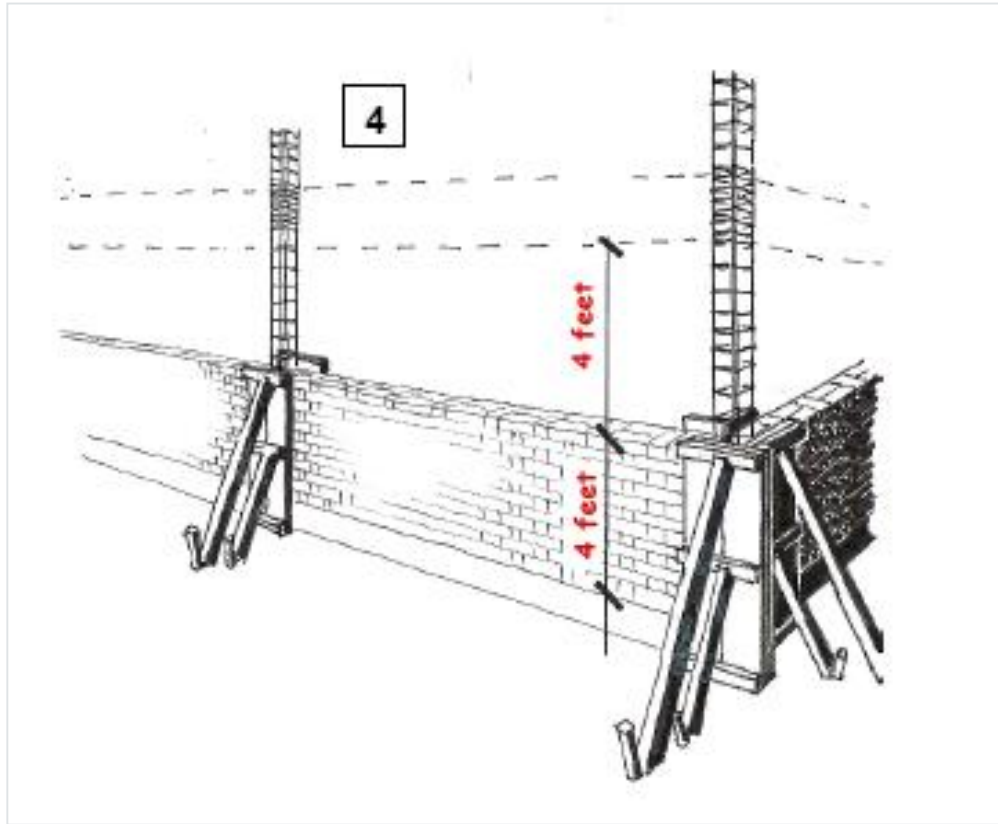
Minimum foundation depth & width 900 mm

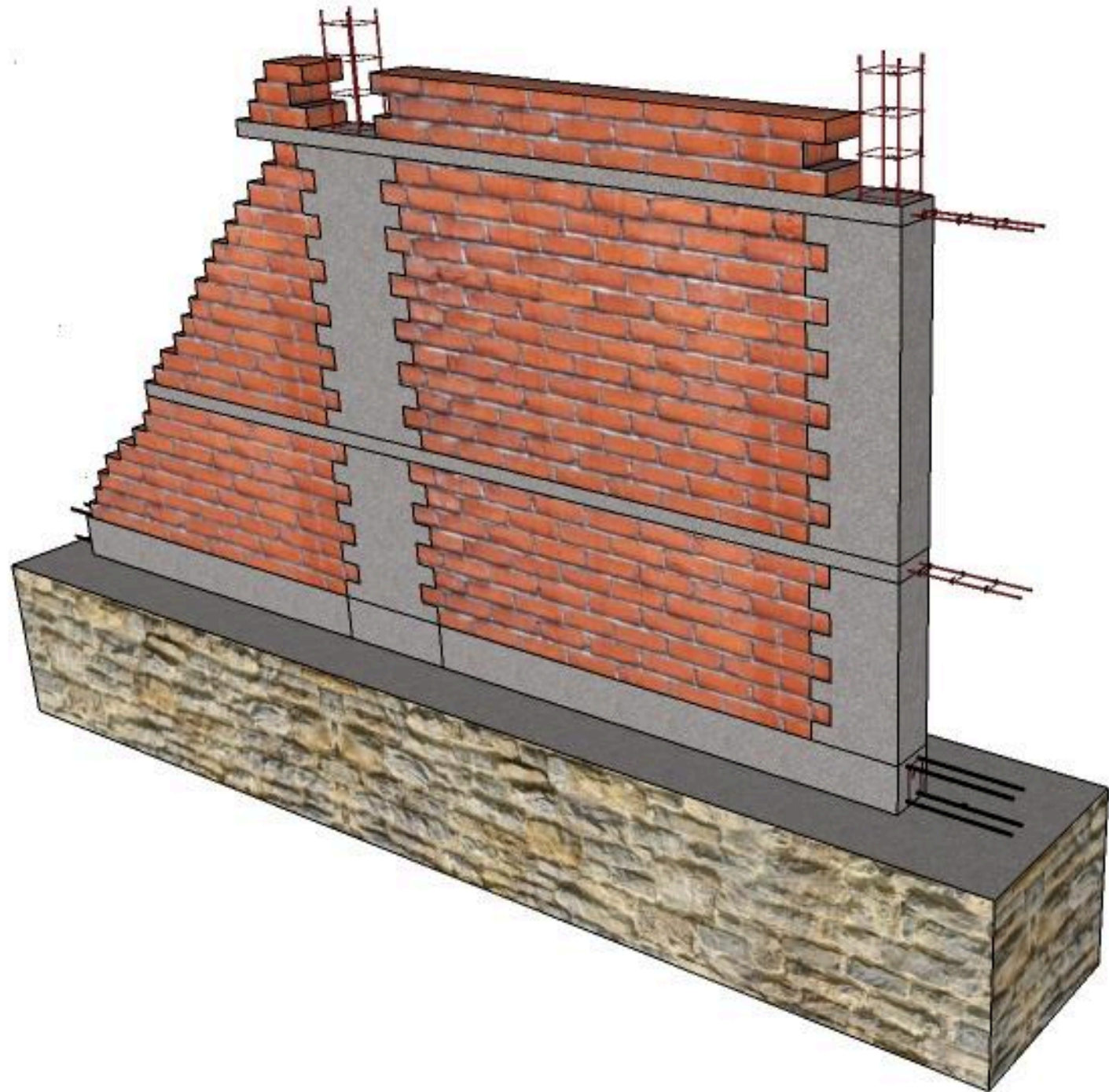
Slab



- Should be on uniform level
- Minimum thickness: 125 mm
- Reinforcement: 8mm bar @ 150mm c/c
- Minimum Cover: 15 mm

Construction Process





Thank You !